

Trace elements in surface water and groundwater in the Surma–Kushiyara Floodplain Basin, Bangladesh: A multivariate statistical evaluation

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ABSTRACT

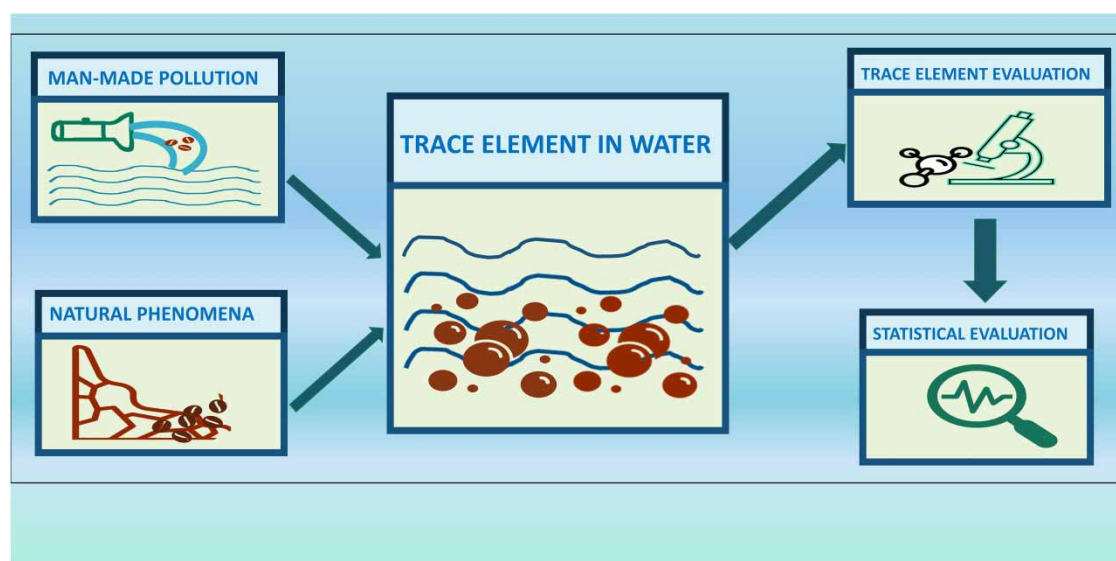
Trace elements are found in small concentrations in water but can be detrimental. Univariate, bivariate, and multivariate analyses of trace elements in the eastern Surma–Kushiyara Floodplain Basin are presented in this study for selected trace elements (As, Co, Cu, Cd, Fe, Zn, Ni, and Mn) and water quality parameters (TDS, pH, and EC). Except for arsenic, manganese, and iron, the research area's trace element concentration of water remains below the drinking water standard. Principal component analysis (PCA) has identified the impact of river inflows, the effect of atmospheric precipitation, biogenic processes, and human activities as possible contributors to the water quality of the study site. Based on their properties, cluster analysis (CA) divided both the surface and groundwater sample points into three major categories. To identify natural links among the water samples, the groups derived from the CA and the natural grouping of surface and groundwater were reassessed based on the discriminant analysis (DA). The classification of surface water and groundwater, and the natural difference between surface and groundwater quality, were strongly supported by the DA. The findings of the study will help policymakers make decisions on safeguarding water and reducing environmental pollution in the study region.

Key words: Bangladesh, groundwater, multivariate analysis, surface water, trace elements, water quality parameters

HIGHLIGHTS

- The quality of surface and groundwater in a floodplain basin can be characterized through the assessment of trace elements in the water.
- Multivariate analysis methods like principal component analysis (PCA), cluster analysis (CA), and discriminant analysis (DA) are used in water quality-related studies to identify the significant contaminant sources of the water samples.
- By recognizing the probable source of water contamination, policymakers can take the appropriate action to prohibit disposal and regulate water consumption.

GRAPHICAL ABSTRACT



1. INTRODUCTION

Water, as a solvent, is vital for living organisms. It is a limited resource used for drinking, agricultural, and industrial purposes. Polluted water is dangerous for plants, animals, and overall biodiversity and can be a vital threat to human health. With the rapid economic and social development of recent decades, pollution sources from livestock, poultry, aquaculture, planting, and rural domestic sewage have drawn much attention. The most prestigious health-related organization in the world, the World Health Organization (WHO), recommends drinking water guidelines rather than binding water quality requirements (Sayato 1989; RS2 2012). As guidelines for appropriate levels of chemical contamination are extensive, the WHO has stated that only the most probable source of risk as determined by experts should be assessed rather than a wide range of chemicals specified in the guideline for drinking water in developing countries like Bangladesh (Gadgil 1998).

Trace elements such as arsenic, cobalt, copper, cadmium, iron, zinc, nickel, and manganese are found in deficient concentrations in the environment. Living organisms require some trace elements in minimal amounts, but high levels of these same elements can be toxic (Obasi & Akudinobi 2020). Iron, for example, is required by many living organisms. Iron transports oxygen throughout the body in human blood. However, consuming too much iron can have adverse effects on human health (Abbaspour *et al.* 2014).

When trace elements are released from rocks, their levels in the environment rise; these releases can occur as a result of natural processes or human activities (Lee & Von Lehmden 2012). Mineral extraction, stormwater runoff, industrial effluents, nuclear reactions, and natural processes like rock weathering, mid-ocean ridge spreading, and volcanic activity all contribute trace elements to the environment (Mohammed *et al.* 2011). In the hydrological cycle, sediments play a critical role as transporters of trace elements. As a part of sustainable development, quantifying pollution levels of trace elements, identifying potential sources, and mitigating future pollution risks are very essential.

Multivariate analysis has been used in situations where several measurements are made on each ongoing item, and the relationships between these measurements and their structures are crucial (Goodman *et al.* 1979). Principal component analysis (PCA), factor analysis (FA), cluster analysis (CA), and discriminant analysis (DA) are practical multivariate statistical approaches for environmental studies that identify hidden relationships belonging to variables and reduce large and complex chemical data sets to a small number of factors, allowing for a better understanding of water quality and potential sources that affect the study region (Ali *et al.* 2016; Wang *et al.* 2017; Hasan *et al.* 2020).

A multivariate analysis is crucial in the meticulous examination of trace elements for presenting a better view of estimation with simplicity (Singh *et al.* 2017). Thuyet *et al.* (2016) did a CA followed by principal components (PCs) of trace elements in shallow groundwater, while Wang *et al.* (2017) evaluated trace elements at three different points in a separate layer of groundwater. Manzoor *et al.* (2006) did a multivariate analysis of trace elements

for textile effluent and found that chromium and lead were dominant. A heavy metal assessment-related study was conducted by [Armah *et al.* \(2010\)](#) in a mining district in Ghana using a multivariate statistical method. Trace elements such as cadmium, copper, arsenic, lead, mercury, zinc, manganese, and iron were used for the study. A water quality assessment study of trace elements was conducted by [Liang *et al.* \(2018\)](#) using the multivariate method. Human health-related analyses were also conducted in that study. The quality of the selected source was found to be excellent. A risk estimation-based study on trace elements was done by [Jakhu & Mehra \(2018\)](#), and the finding is that the water is not suitable for drinking purposes as it has carcinogenic action. In addition, [Siepak & Sojka \(2017\)](#) introduced DA along with the principal component and CA for surface water, while [Kumar *et al.* \(2020\)](#) analyzed trace elements for three different rivers.

As a result of the researcher's awareness of the importance of trace metal analysis, the analysis of trace elements has become a frequent topic. A comprehensive study on trace elements in surface water was done by [Khound & Bhattacharyya \(2017\)](#) with seasonal variation in two meteorological years (2009–2011). There were information limitations in this study due to the methods of multivariate analysis. However, other statistical estimation approaches revealed actual trends. Trace metal analysis-related studies either on groundwater or surface water are quite popular among researchers, but a study of both groundwater and surface water in terms of trace elements is hard to find. A study conducted by [Krishna *et al.* \(2009\)](#) gives a better view of heavy metal analysis by taking a sample from ground and surface water sources. However, the collection point for surface water in the study region is not the same as the groundwater sample. Taking groundwater and surface water samples from similar collection points may give a better understanding of polluted water chemistry, as toxic pollutants from the surface can move through the soil and end up contaminating the groundwater ([Sophocleous 2002](#); [Song *et al.* 2006](#)).

Although Bangladesh has abundant water resources, due to its dense population and ineffective control and management of water resources, water pollution has become a serious concern for sustainable development in the country ([Hasan *et al.* 2019](#)). This study, therefore, aims to assess trace elements and water quality parameters in surface and groundwater in the first part of the Eastern Surma–Kushiyara Floodplain Basin using the required descriptive and multivariate statistical methods to help policymakers to make better decisions on safeguarding water and reducing the environmental pollution in the study region.

2. METHODS

2.1. Study area

Sylhet City, one of Bangladesh's fastest-growing cities, is in an area prone to flash floods. Adjacent to the city, the Eastern Surma–Kushiyara Floodplain Basin Region is located in the relatively higher sections of the Surma–Kushiyara (rivers) floodplain. This region has been created by river sediments draining into the Meghna catchment area from the northern and eastern hills. Depending on the geological formation, the water quality varies from place to place. Alluvial silt and clay dominate the surface geology of the region ([Shamsudduha *et al.* 2011](#)). This region covers 4 km² and includes the districts of Sylhet, Moulavi Bazar, Sunamganj, and Habiganj. Water quality has deteriorated due to rapid population growth, and waste product expansion is producing other hazardous issues. Groundwater is the primary source of drinking water and other domestic work for the locals, while surface water is used for irrigation and other purposes.

The study area spans around 250 km². It is located between the longitudes of 91,055'0" E and 92,006'0" E and the latitudes of 24,043'0" N and 24,054'0" N in the first part of the Eastern Surma–Kushiyara Floodplain. It is about 8 km east of Sylhet City. This region is bounded on the north by the Surma River and the south by the Kushiyara River, and it is located in the center of Gulapgonj Upazila ([Figure 1](#)). The research area was chosen because the water quality of the Surma and Kushiyara rivers has an impact on the study area, and there has not been any comprehensive and organized water quality study here.

2.2. Sampling and testing

Four water samples were collected from each of the seven unions and one municipality within the selected area (one sample for surface water and another for groundwater from an area of four adjacent wards). Thus, the total number of samples is 32 for surface water and 32 for groundwater. The general approach to selecting the sampling station for the study was to select a location with both surface and groundwater sources. As groundwater sources are widespread in the study area as a key source for drinking and domestic work, we initially chose a surface water source (pond, lake, canal, etc.) for collecting samples. Local people's assistance and

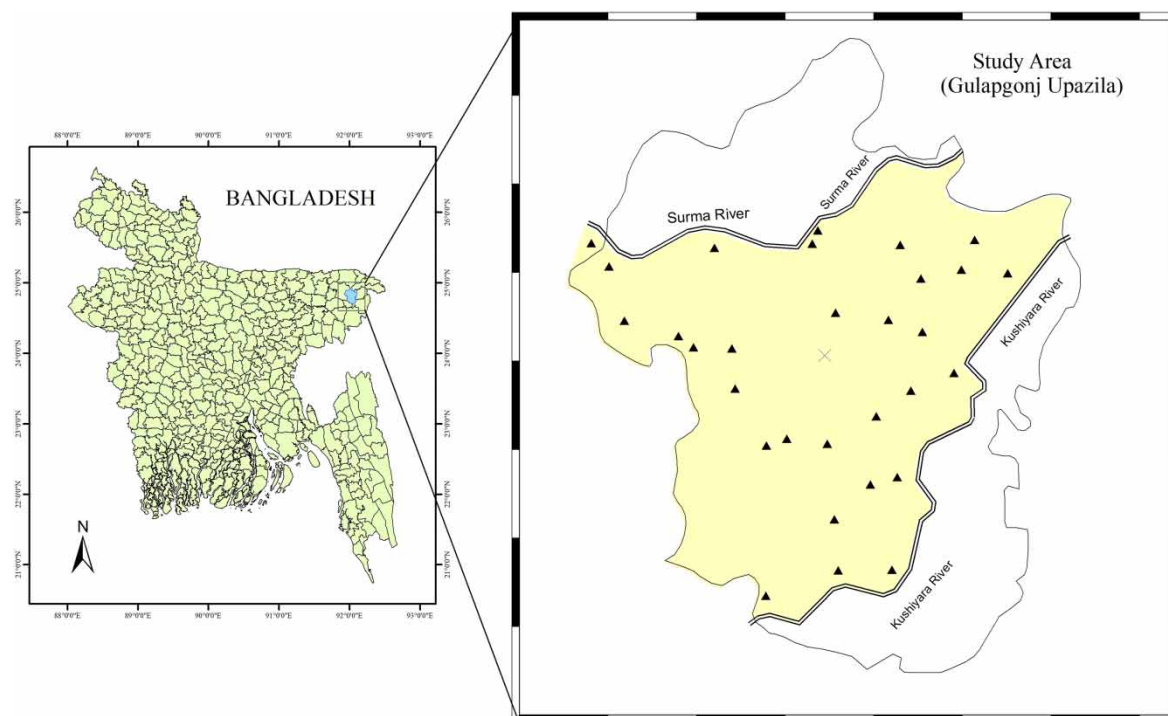


Figure 1 | A study area map.

guidance were very helpful in identifying surface water sources. To avoid exposure to direct sunlight, samples of surface water were collected from a depth of about 1 ft (30.5 cm) below the surface. After five pumps, groundwater samples were taken from the tube well. Unfortunately, due to our failure to enforce the considerations, two samples were not taken for analysis. Therefore, the total number of samples is 30 for surface water and 30 for groundwater. Each site's longitude and latitude were taken from Google Maps, and the site locations were recorded. The sampling sites for the study area are shown in Table 1.

Physical parameters (pH, EC, and TDS) were measured using the HI9813-6 Portable pH/EC/TDS/Temperature Meter while collecting samples, and the collected water was stored at pH < 2 (using concentrated HNO₃) and under 4 °C in precleaned high-density polyethylene (HDPE) bottles. The testing equipment was carried out to test water quality parameters straight from the source during the sample collection process. Before collecting samples, the testing kit was properly calibrated, and after each test, the testing kit was perfectly cleaned with distilled water to improve the results.

By using the ICP-OES (inductively coupled plasma-optical emission spectrometry) technique and the Avio 200 ICP-OES equipment, trace elements such as, Co, Cu, Cd, Mn, Fe, Zn, and Ni were investigated in the laboratory with detection limits of 1, 0.14, 0.26, 0.1, 0.026, 0.08, 0.07, and 0.4 µg/L, respectively.

2.3. Statistical analysis

To analyze complex data sets with several variables, in addition to employing descriptive statistics, multivariate analysis methods such as PCA, CA, and DA were applied. These techniques can reduce the number of dimensions and provide useful underlying information from the original data. The study also employed correlation analysis, such as Pearson's correlation and partial correlation analysis, to demonstrate the inter-relationship between selected parameters. All statistical analysis was performed using IBM SPSS Statistics V23.0.

3. RESULTS AND DISCUSSION

3.1. Descriptive statistics

A trace metal analysis of collected surface water and groundwater with their descriptive statistics is summarized in Table 2. In addition, Table 3 shows a comparison with recommended levels specified by the WHO and Bangladeshi standards (BS) suggested by the Department of Public Health Engineering (DPHE) (2021).

Table 1 | Geographic location of the sampling site points

ID	Location name	Longitude	Latitude
1	Hawortola	24.7284	91.9945
2	Guashpur	24.7378	92.0401
3	Shekhpur	24.7376	92.0207
4	Vadeyshor	24.7561	92.0193
5	Ujan Meherpur	24.7714	92.042
6	Silimpur	24.7688	92.0323
7	Kanishail	24.7935	92.0345
8	Korgoan	24.7835	92.0167
9	Nowai	24.7854	92.002
10	Fulshaind	24.7829	91.9946
11	Lokkhanaband	24.8035	91.9833
12	Jhangalhata	24.8182	91.9821
13	Laxmipasha	24.815	91.9682
14	Nimadol	24.8227	91.9628
15	Dora	24.8281	91.9431
16	Rofipur	24.8479	91.9376
17	Barocut	24.831	92.0198
18	Dharabor	24.8285	92.0389
19	Shilghat	24.8241	92.0512
20	Nagar	24.8028	92.047
21	Sunampur	24.8093	92.0626
22	Amnia	24.8435	92.0506
23	Amora	24.8467	92.0653
24	Sundishail	24.8576	92.0701
25	Gagua	24.8454	92.0821
26	Ghugarkul	24.8557	92.0431
27	Saraswati	24.8609	92.0133
28	Kadamgach (Bazar)	24.8561	92.0113
29	Hazipur	24.8547	91.9758
30	Pachmail	24.8563	91.9311

According to [Tables 2 and 3](#), the concentration of arsenic levels in surface water ranges from 0 to 85 $\mu\text{g/L}$, with a mean value of 42.4 $\mu\text{g/L}$, exceeding the WHO recommended level of 10 $\mu\text{g/L}$. However, in terms of BS of 50 $\mu\text{g/L}$, this level is acceptable. Nonetheless, for the groundwater sample, the mean value of arsenic is 51.867 $\mu\text{g/L}$, and it exceeds both the recommended levels from the WHO and BS. Most of the time, a high concentration of arsenic in natural water is caused by alluvium or loess ([Smedley & Kinniburgh 2002](#)). The mean concentrations of cobalt are 2.5 $\mu\text{g/L}$ for groundwater and 4.07 $\mu\text{g/L}$ for surface water. Neither the WHO nor the BS recommends a value for cobalt. So, we may take the prevailing levels as acceptable for usage. The WHO recommends a maximum copper concentration of 2000 $\mu\text{g/L}$ and the BS recommends a concentration of 1000 $\mu\text{g/L}$. In contrast, our observed maximum concentration levels of copper are 22 and 5 $\mu\text{g/L}$ for surface water and groundwater, respectively. As a result, the concentration of copper is not a concern in the study region. The maximum cadmium concentration for both ground and surface water is 2 $\mu\text{g/L}$. It is within the acceptable limit since it is suggested by the WHO as 3 $\mu\text{g/L}$ and by the BS as 5 $\mu\text{g/L}$. The suggested level of manganese by both the WHO and BS is 100 $\mu\text{g/L}$. Our observed mean values of manganese are 74.767 $\mu\text{g/L}$ for groundwater and 143.3667 $\mu\text{g/L}$ for surface water. A significant number of surface water samples exceed the recommended level. Manganese can stay in groundwater or surface water in some acidic or anaerobic conditions ([RS2 2012](#)). However, some compounds based on manganese are hazardous and cause cancer ([Gerber *et al.* 2002](#)).

Table 2 | Statistical summary of targeted trace metal concentrations in the sample of surface water and groundwater

Parameters	Surface water					Groundwater				
	Mean	SD	Max	Min	CV%	Mean	SD	Max	Min	CV%
As	42.4	31.02	85	0	73.16	51.87	29.38	96	0	56.64
Co	4.07	4.24	17	0	104.31	2.5	2.3	5	0	92.08
Cu	3.4	5.85	22	0	171.95	1.4	1.77	5	0	126.67
Cd	0.93	0.94	2	0	101.19	0.87	0.9	2	0	103.8
Mn	143.37	154.82	540	2	107.99	74.77	76.2	290	0	101.9
Fe	1879.07	3390.05	14,046	0	180.41	566.73	942.59	4,786	0	166.32
Zn	18.2	50.28	270	0	276.25	0.07	0.37	2	0	547.72
Ni	12.37	10.13	32	0	81.93	10.77	10.4	23	0	96.61
pH	7	0.66	8.4	5.7	9.45	5.97	0.96	7.6	4.1	16.07
EC	0.14	0.8	0.29	0.04	58.72	0.18	0.14	0.54	0.01	79.06
TDS	103.47	55.61	209	33	53.74	133.7	97.01	389	22	72.74

Note: All values are in µg/L, except pH, EC (mS/cm), and TDS (mg/L).
SD, standard deviation; CV, coefficient of variation.

Table 3 | Comparison of standard specifications of trace elements for groundwater and surface water

Parameters	Mean (surface water)	Mean (groundwater)	WHO	BS
As	42.4	51.87	10	50
Co	4.07	2.5	–	–
Cu	3.4	1.4	2000	1000
Cd	0.93	0.87	3	5
Mn	143.37	74.77	100	100
Fe	1879.07	566.73	–	1000
Zn	18.2	0.07	–	5000
Ni	12.37	10.77	20	100
pH	7	5.97	6.5–8.5	–
EC	0.13	0.18	–	–
TDS	103.47	133.37	1000	1000

Note: All values are in µg/L, except pH, EC (mS/cm), and TDS (mg/L).

WHO, World Health Organization; BS, Bangladeshi standard (recommended by the Department of Public Health Engineering, Bangladesh).

The mean concentration of iron for surface water is 1879 µg/L and the maximum concentration is 14,046 µg/L. Again, the mean concentration of the same ion for groundwater is 566 µg/L with the maximum value being 4786 µg/L. The recommended level is 1000 µg/L by BS and, therefore, the concentration of iron in the surface water is high. Iron is an essential mineral as it assists blood flow, so the concentration found is not considered a health hazard. However, highly concentrated water with iron may increase turbidity and turn reddish-brown. The concentrations of zinc in our collected samples are not concerning at all. The Bangladeshi standard recommends a zinc concentration of 5000 µg/L. The maximum concentration of our sample is 270 µg/L for surface water and 2 µg/L for groundwater. The mean concentration of nickel in our water sample is within the limit of 20 µg/L by Bangladeshi standards and 100 µg/L by the WHO. It is to be noted that a high concentration of nickel has a carcinogenic mechanism and may cause skin problems (Obasi & Akudinobi 2020).

Water quality parameters such as pH, EC, and TDS do not exceed the suggested levels. pH levels of most of the samples are within the range of 6.5–8.5. However, some samples seem to be acidic as low pH values have been observed. Therefore, from the comparison of the amounts of particular trace metal elements and water quality parameters that are present in the surface and groundwater at the site area with WHO- and BS-suggested

levels, we may conclude that the quality of both the surface and groundwater in this area is suitable for domestic, drinking, agricultural, and industrial purposes.

3.2. Correlation analysis

For the surface water samples, out of 55 pairs of water quality attributes, 15 pairs of the samples have been found to be significantly correlated ($p < 0.05$), which are marked as underlined in Table 4.

Table 4 | Correlation coefficients in surface water samples

Parameters	As	Co	Cu	Cd	Mn	Fe	Zn	Ni	pH	EC	TDS
As	1.00										
Co	0.16	1.00									
Cu	0.14	0.09	1.00								
Cd	<u>0.40</u>	<u>0.48</u>	0.01	1.00							
Mn	−0.47	−0.17	−0.29	−0.28	1.00						
Fe	− <u>0.53</u>	− <u>0.38</u>	−0.29	− <u>0.45</u>	<u>0.49</u>	1.00					
Zn	−0.13	0.32	0.05	0.10	0.40	−0.11	1.00				
Ni	<u>0.50</u>	<u>0.71</u>	−0.01	<u>0.80</u>	−0.18	− <u>0.42</u>	0.33	1.00			
pH	−0.13	−0.35	−0.18	0.12	0.04	0.22	0.02	−0.03	1.00		
EC	0.14	0.03	−0.34	<u>0.39</u>	0.19	−0.08	0.19	0.28	0.34	<u>1.00</u>	
TDS	0.14	0.02	−0.34	<u>0.38</u>	0.19	−0.07	0.19	0.27	0.34	<u>1.00</u>	1.00

There is a perfect positive relationship between EC and TDS ($r = 1$) because TDS-EC has an algebraic relationship. While nickel has strong positive relationships with cobalt ($r = 0.71$) and cadmium ($r = 0.80$), it also has a moderate positive correlation with arsenic ($r = 0.50$) and a moderate negative correlation with iron ($r = -0.42$). Iron has a moderately negative correlation with arsenic ($r = -0.53$), cobalt ($r = -0.38$), and cadmium ($r = -0.45$) separately, but it is moderately positively related with manganese ($r = 0.49$). Cadmium has a positive correlation with arsenic ($r = 0.40$) and copper ($r = 0.48$). In addition, cadmium has a notable positive correlation with EC or TDS. Zinc has a moderately positive correlation with manganese ($r = 0.40$), but manganese is moderately negatively related to arsenic ($r = -0.47$).

In Table 5, the partial correlation coefficients are presented. It shows that, when the effects of other elements are controlled, only the correlations of nickel with arsenic, cobalt, and cadmium and EC with TDS remain significant at a 5% level of significance with the signs as before. In addition, the correlations of cadmium with arsenic and pH are negatively significant. All these suggest that it may be the case that among the shallow soil-forming metals and water flows, nickel and cobalt mainly dominate the surface water quality of the site area.

Table 5 | Partial correlation coefficients among the elements in surface water

Parameters	As	Co	Cu	Cd	Mn	Fe	Zn	Ni	pH	EC	TDS
As	1.00										
Co	− <u>0.52</u>	1.00									
Cu	0.11	0.03	1.00								
Cd	−0.35	−0.20	0.12	1.00							
Mn	−0.28	−0.24	−0.14	−0.12	1.00						
Fe	−0.34	−0.16	−0.12	−0.29	0.34	1.00					
Zn	−0.28	0.01	0.23	−0.34	0.41	−0.33	1.00				
Ni	<u>0.66</u>	<u>0.66</u>	−0.16	<u>0.70</u>	0.14	0.22	0.39	1.00			
pH	−0.30	− <u>0.49</u>	−0.01	0.10	−0.30	0.18	0.12	0.23	1.00		
EC	−0.12	−0.05	0.01	0.05	−0.06	−0.05	−0.03	0.11	0.00	1.00	
TDS	0.14	0.05	−0.03	0.00	0.07	0.04	0.04	−0.12	0.02	<u>1.00</u>	1.00

Again, for groundwater samples, out of 55 pairs of water quality attributes, 25 pairs of the collected samples are found to be significantly correlated ($p < 0.05$), and they are distinguished with underlines in Table 6.

Table 6 | The values of the correlation coefficient in the sample of groundwater

Parameters	As	Co	Cu	Cd	Mn	Fe	Zn	Ni	pH	EC	TDS
As	1.00										
Co	<u>0.51</u>	1.00									
Cu	<u>0.75</u>	<u>0.68</u>	1.00								
Cd	<u>0.46</u>	<u>0.93</u>	<u>0.51</u>	1.00							
Mn	−0.59	−0.78	−0.69	−0.75	1.00						
Fe	−0.27	−0.55	−0.44	−0.49	<u>0.46</u>	1.00					
Zn	0.03	−0.21	−0.15	−0.18	<u>0.36</u>	−0.09	1.00				
Ni	<u>0.69</u>	<u>0.94</u>	<u>0.85</u>	<u>0.86</u>	−0.83	−0.54	−0.20	1.00			
pH	−0.18	0.08	−0.03	0.10	0.09	−0.44	0.07	−0.01	1.00		
EC	−0.34	−0.11	−0.21	−0.14	0.19	−0.05	−0.04	−0.22	<u>0.64</u>	1.00	
TDS	−0.33	−0.11	−0.21	−0.12	0.19	−0.05	−0.04	−0.22	<u>0.64</u>	<u>1.00</u>	1.00

From this correlation coefficient table, we have learned that TDS and EC have a perfect positive correlation ($r = 1$), and it should be so. Besides, pH is moderately positively correlated with TDS or EC ($r = 0.64$), but moderately negatively correlated with iron ($r = -0.44$). Nickel has a strong positive correlation with arsenic ($r = 0.69$), cobalt ($r = 0.94$), copper ($r = 0.85$), and cadmium ($r = 0.86$), but a strong negative correlation with manganese ($r = -0.83$) and a moderately negative correlation with iron ($r = -0.54$) on its own. Likewise, iron has a moderate positive correlation with manganese ($r = 0.46$) but a moderate negative correlation with cobalt ($r = -0.55$), copper ($r = -0.44$), and cadmium ($r = -0.49$). Manganese has only a negative correlation with arsenic ($r = -0.59$), cobalt ($r = -0.78$), copper ($r = -0.69$), and cadmium ($r = -0.75$). Cadmium has a strong positive stimulus with cobalt ($r = 0.93$) and moderately significant correlations with arsenic ($r = 0.46$) and copper ($r = 0.51$). Copper has a strong positive correlation with arsenic ($r = 0.75$) and cobalt ($r = 0.68$). Finally, cobalt and arsenic ($r = 0.51$) and zinc and manganese ($r = 0.36$) have a moderate positive correlation among themselves.

All the significant partial correlation coefficients that are generated from the groundwater elements, as underlined in Table 7, are supported by the corresponding Karl Pearson coefficients in Table 6. The correlation between copper and cadmium was converted from positive to negative. It is indicated that soil, rock-forming minerals, and water levels influence the quality of the groundwater in the site area.

Table 7 | Partial correlation coefficients among the elements in groundwater

Parameters	As	Co	Cu	Cd	Mn	Fe	Zn	Ni	pH	EC	TDS
As	1.00										
Co	−0.23	1.00									
Cu	0.26	−0.19	1.00								
Cd	0.11	<u>0.48</u>	−0.57	1.00							
Mn	−0.17	0.14	−0.03	−0.26	1.00						
Fe	0.15	−0.09	0.09	0.21	0.30	1.00					
Zn	0.31	−0.06	−0.06	0.09	<u>0.43</u>	−0.33	1.00				
Ni	0.15	<u>0.61</u>	<u>0.78</u>	0.32	−0.14	−0.15	0.01	1.00			
pH	−0.02	−0.08	0.25	0.36	0.31	−0.56	−0.10	−0.19	1.00		
EC	−0.11	0.37	0.29	−0.13	−0.12	0.09	0.12	−0.33	0.12	1.00	
TDS	0.10	−0.35	−0.29	0.11	0.10	−0.06	−0.12	0.33	−0.06	<u>1.00</u>	1.00

From the results in Tables 4 and 6 of correlation coefficients based on the data generated from the collected sample of surface and groundwater, it is apparent that among the common significant relationships in types of water, the signs (positive or negative) coincide. Except for the relationships between manganese and iron and zinc, and between EC and TDS, the sizes of the correlation coefficient values are smaller for surface water. In other words, the strength of relationships is greater in groundwater. Besides, the relationships of copper with arsenic, cobalt, cadmium, manganese, iron, and nickel; manganese with cobalt and cadmium; nickel with manganese; cobalt with arsenic; pH with manganese; EC with manganese; or TDS with manganese are significant only for groundwater.

3.3. Principal component analysis

For surface water samples, the PCA along with the rotation method of Varimax with Kaiser Normalization extracted three significant PCs with eigenvalues > 1 and is presented in Table 8. The extracted PCs cover almost 70% of the total variance in the corresponding surface water quality data set on trace elements. The screen plot curve shown in Figure 2 gives a clear visualization of the associated variance with each factor, where the steep slope shows the biggest factors. The figure clearly shows that there are three dominant factors in the total variance of the parameters of water. Figure 3 is a radar chart and shows the loadings of the trace elements for three dominant factors. Changes in the loadings of elements to factors are also visually depicted. It is clear from the figure that the third factor (PC3) is more overlaid on the first factor (PC1) than the second factor (PC2).

Table 8 | The rotated factor loadings for the samples of surface water

Parameters	PC1	PC2	PC3
As	0.789	0.061	0.066
Co	0.284	-0.179	0.804
Cu	0.271	-0.534	0.046
Cd	0.616	0.349	0.482
Mn	-0.812	0.246	0.238
Fe	-0.714	0.12	-0.32
Zn	-0.334	0.094	0.748
Ni	0.519	0.193	0.727
pH	-0.049	0.602	-0.3
EC	0.098	0.916	0.197
TDS	0.092	0.917	0.191
Eigenvalue	3.424	2.632	1.572
Variance explained (%)	31.13	23.93	14.3
Cumulative variance (%)	31.13	55.06	69.36

Here, PC1 accumulates 31.2% of the total variance and has moderate-to-strong positive loadings on nickel, cadmium, and arsenic and nominal positive loadings on cobalt, copper, and EC or TDS. It also has strong negative loadings on iron and manganese and insignificant negative loadings on zinc and pH. As arsenic is positively correlated with cadmium and nickel, and each of them is again negatively correlated with manganese and iron (Table 4), PC1 is influenced by the issues that are dominated by arsenic, cadmium, and nickel and reduce the presence of iron and manganese. This contamination's probable sources include industrial processes, dietary sources, fertilizers, sewage sludge, and rock weathering. So, a likely cause of this could be the inflows of contamination from the Surma and Koshiyara rivers. As a result, this PC could be referred to as 'the influence of river inflows.'

For PC2, in addition to a moderate positive loading on cadmium, this PC has strong positive loadings on pH, EC, or TDS. It accounts for 23.93% of the total variance. Since cadmium and pH, EC, or TDS have a positive correlation (Table 3), the component might only be affected by physical causes. Therefore, rain, geology, and

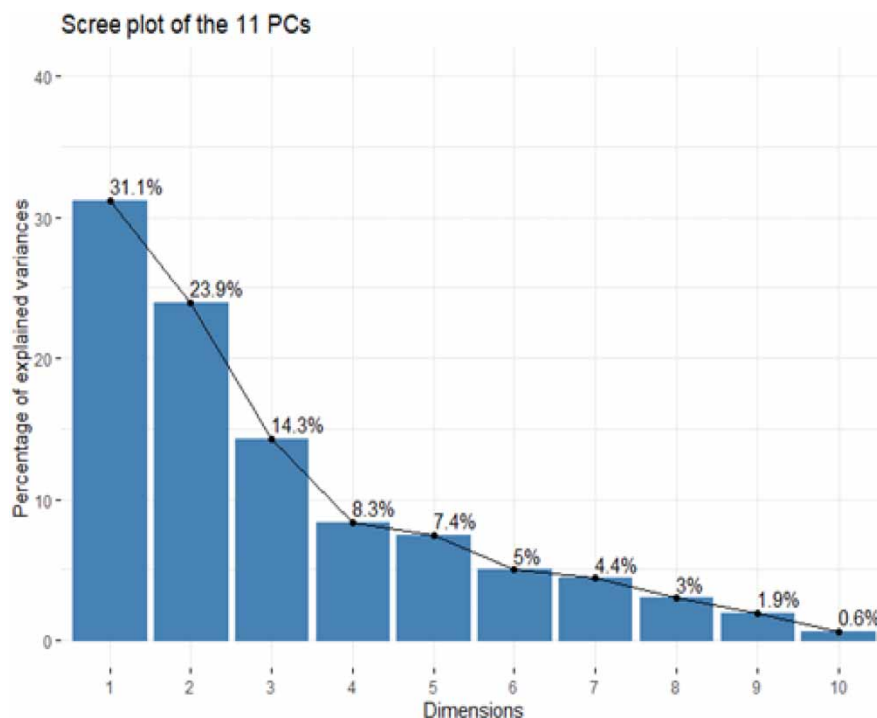


Figure 2 | A surface water scree plot with percentage (%) of explained variance.

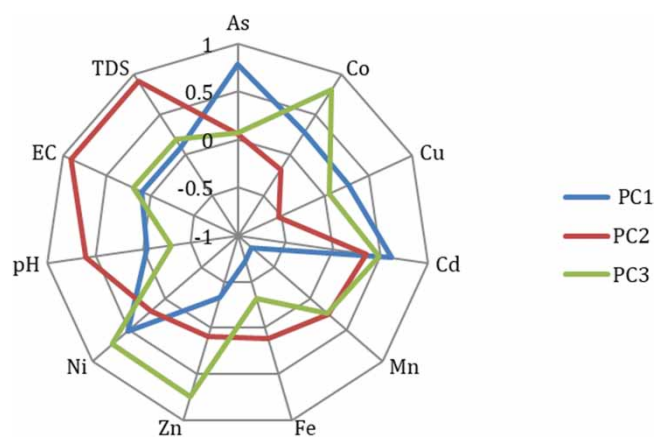


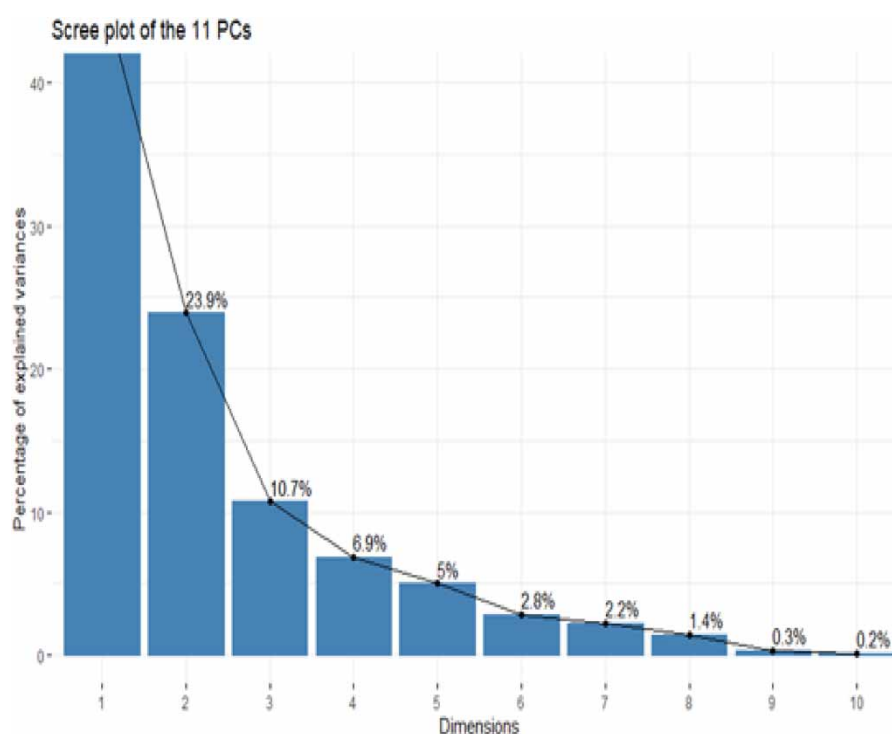
Figure 3 | Rotated factor loadings of three surface water factors extracted by PCA.

evaporation are the main variables that could have an impact on the presence of these interactions. As a result, this PC might be leveled as ‘the effect of atmospheric precipitation.’ Finally, cobalt, zinc, nickel, and cadmium dominate PC3, and they can be attained in the water from biogenic substances. This PC3 accumulates 14.3% of the total variance and also has nominal negative loadings for pH and iron. Accordingly, it is indicated that this factor is controlled by ‘the biogenic processes.’ It is to be noted that the biogenic processes also affect the water quality of the Surma and Kushiya rivers, and as a result, this factor is more overlaid on the factor of river inflows (Figure 3).

For groundwater samples, PCA along with the rotation method of Varimax with Kaiser Normalization extracted three significant PCs with eigenvalues >1 and are presented in Table 9. These three factors cover about 81.21% of the total variance in the corresponding groundwater quality data set on trace elements. Figure 4 illustrates a screen plot curve that clearly shows the related variation associated with each factor. The steepest slope indicates the most significant factors. The figure clearly shows that three dominant factors are enough to explain the total variance of the selected attributes of groundwater.

Table 9 | The rotated factor loadings for the samples of groundwater

Parameters	PC1	PC2	PC3
As	0.698	−0.336	0.152
Co	0.933	0.037	−0.112
Cu	0.828	−0.148	0.017
Cd	0.868	0.032	−0.115
Mn	−0.864	0.107	0.277
Fe	−0.663	−0.311	−0.384
Zn	−0.169	−0.039	0.916
Ni	0.974	−0.098	−0.068
pH	0.111	0.848	0.234
EC	−0.159	0.934	−0.103
TDS	−0.154	0.933	−0.109
Eigenvalue	5.126516103	2.626715060	1.179583089
Variance explained (%)	46.61	23.88	10.72
Cumulative variance (%)	46.61	70.48	81.21

**Figure 4** | A groundwater scree plot with percentage (%) of explained variance.

The radar chart, as shown in Figure 5, shows the loadings of the trace elements of three dominant factors. Changes in the loading of elements for factors are also visually depicted. It is clear from the figure that the third factor (PC3) is more overlaid on the second factor (PC2) than the first factor (PC1).

The PC1 for groundwater samples accumulates 46.61% of the total variance. Except for the loading of zinc in PC3 for surface water, this PC bears all the significant loadings with the same signs as PC1 and PC3 for surface water. Consequently, the sources of ‘the impact of river-inflows’ from Surma and Kushiya and ‘the biogenic processes’ are associated with this factor PC1. Likewise, comparing the loadings of PC2 for groundwater with the loadings of PC2 for surface water, it is not imprudent to conclude that the PC2 for groundwater could be

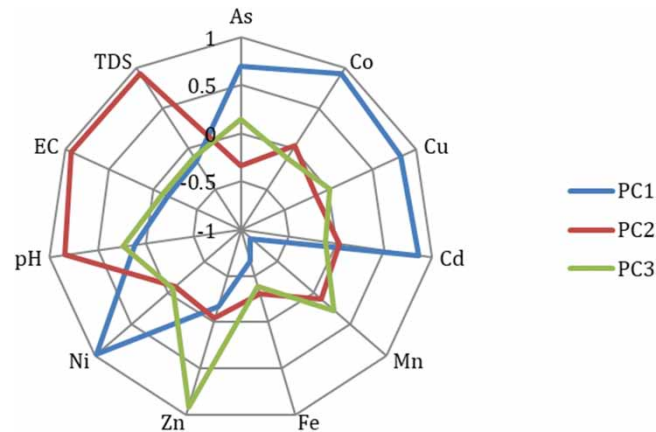


Figure 5 | Rotated factor loadings of three groundwater factors extracted by PCA.

linked with ‘the effect of atmospheric precipitation.’ Finally, as zinc is the only element that is strongly correlated with PC3 for groundwater and this PC3 is overlaid on PC2, we may assume that this PC3 is the result of both the ‘natural processes and human activities’ factor, namely, the use of fertilizers.

3.4. Cluster analysis

As a multivariate exploratory analysis, CA is applied in water quality assessment to identify homogeneous groups of sample points having similar qualities in the parameters. We used the hierarchical agglomerative clustering method in this study, which generates a series of models with cluster solutions ranging from 1 (all sample points in one cluster) to n (each sample point is an individual cluster). To merge the clusters, Pearson correlation-based distance has been used as a measure of dissimilarity. In addition, the average distance of the observations of all data, that is, the between-groups linkage has been utilized as a function of the pairwise distances of observations in the clusters, and thus it specifies the dissimilarity of clusters.

Table 10 shows surface water clusters formed into three major groups, and these are presented in a dendrogram in Figure 6. Group A includes the sample points of water in which arsenic, copper, and cadmium are fully absent and EC or TDS are relatively low. In Group B, although one or more elements of arsenic, copper, and cadmium are present, EC or TDS are still relatively low. Finally, Group C is comprised of the sample points with relatively higher values of EC or TDS. In addition, the sample points of Group A belong to the western part of the site area, whereas except for the point at Rofipur (16), all points of Group B are in the midpart between the Surama and Kushiya rivers. Excluding the point of Jhangalhata (12), all points of Group C were laid near either of the rivers (Figure 7).

Table 10 | Surface water clusters

Group	Location/sample no.	N (%)
A	4, 9, 14, 29,	4 (13.33)
B	11, 13, 16, 17, 18, 19, 23, 24, 26	9 (30.00)
C	1, 2, 3, 5, 6, 7, 8, 10, 12, 15, 20, 21, 22, 25, 27, 28, 30	17 (56.67)

Similar to surface water, for groundwater, the clusters based on the chosen trace elements and water quality parameters are provided in Table 11, and the resultant dendrogram is shown in Figure 8. Group A contains only one sample point, Guashpur (2), with the lowest TDS. Following the TDS of Group A, Group B is comprised of the sample points with lower values of TDS. However, the remaining points belong to the cluster of Group C and have relatively higher values of TDS. Group A is situated at the extreme point of the site area at the northwest corner of the site. Like Group B of surface water, Group B of groundwater is also located in the middle part of the site. The rest of the sample points belong to Group C, and most of these are close to either of the rivers on the south or north sides (Figure 9).

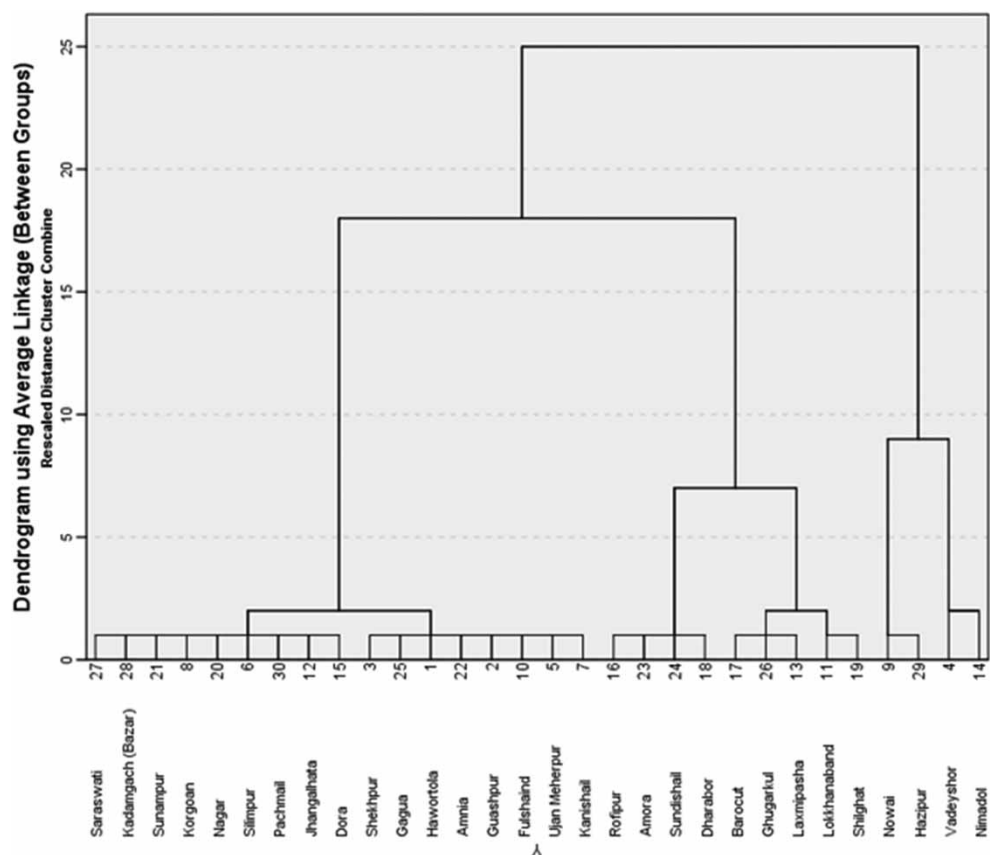


Figure 6 | A dendrogram of CA based on surface water.

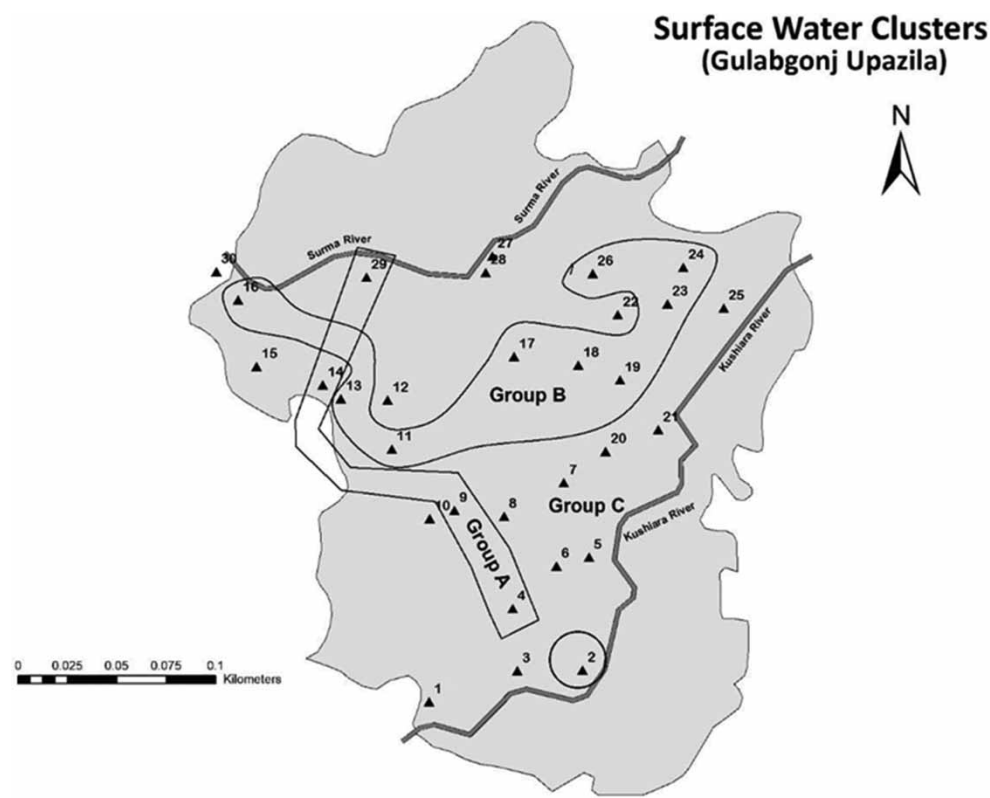
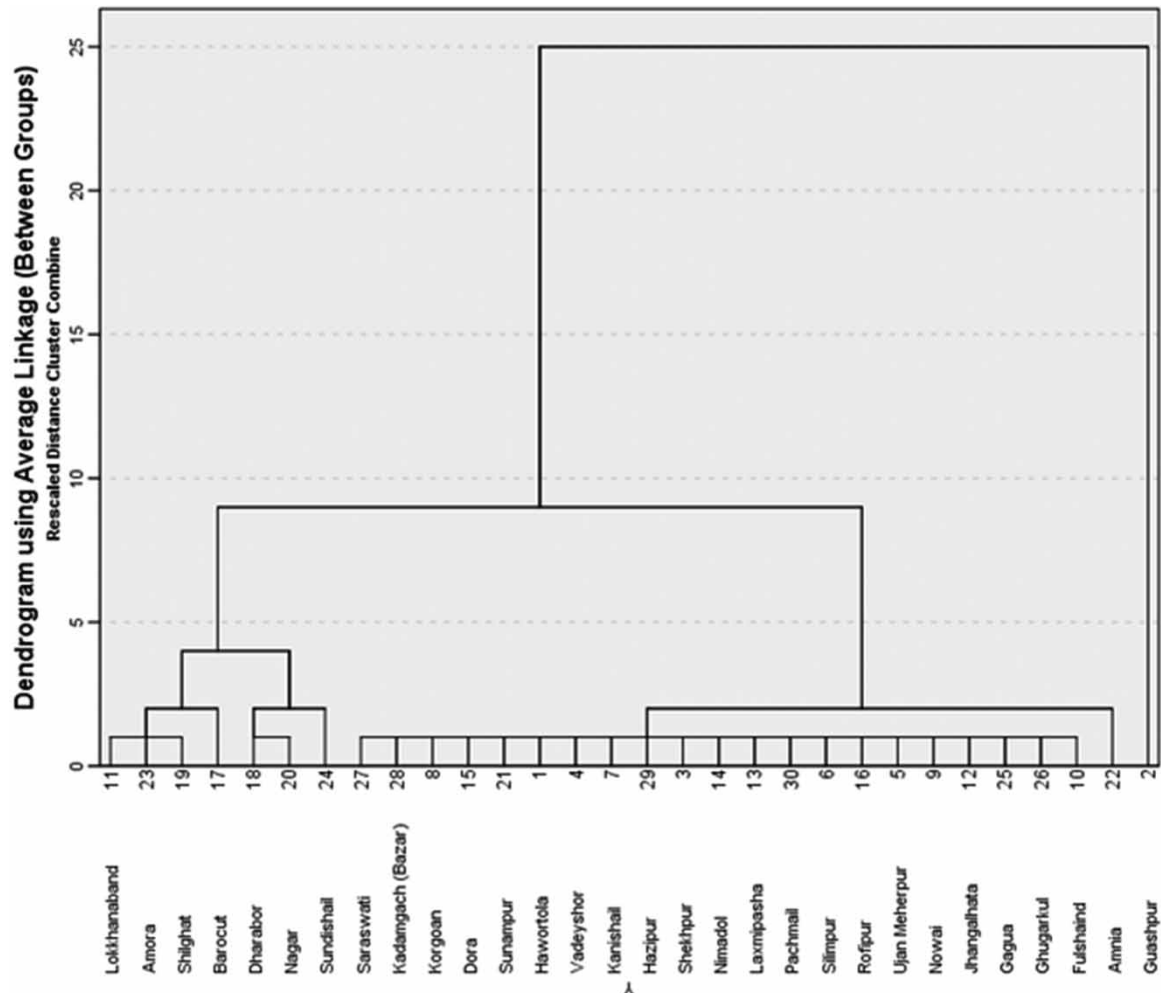


Figure 7 | Clusters of surface water sample points at the site area.

Table 11 | Groundwater clusters

Group	Location/sample no.	N (%)
A	2	1(3.33)
B	11, 17, 18, 19, 20, 23, 24	7(23.33)
C	1, 3, 4, 6, 5, 7, 8, 9, 10, 12, 13, 14, 15, 16, 21, 22, 25, 26, 27, 28, 29, 30	22(73.34)

**Figure 8** | A dendrogram of CA based on groundwater.

The PCA in Section 3.3 alluded to the fact that the inflows from the rivers are the prominent factors that influence the quality of both surface and groundwater. Because the largest cluster of sample points for both surface and groundwater is close to either river, the findings of the CA support those of the PCA.

3.5. Discriminant analysis

DA was done with the expectation that it would help to find the clusters or groups of water sample points that are naturally banded together. The analysis was accomplished by finding discriminant linear functions of trace elements and water quality parameters that help in grouping the water sample points into homogeneous clusters. The three groups obtained from the CA of surface water as well as the three groups obtained from that of groundwater have been selected for this purpose. In addition, two groups that consist of all sample points of surface water and groundwater separately have also been selected. The standard method (based on all studied trace

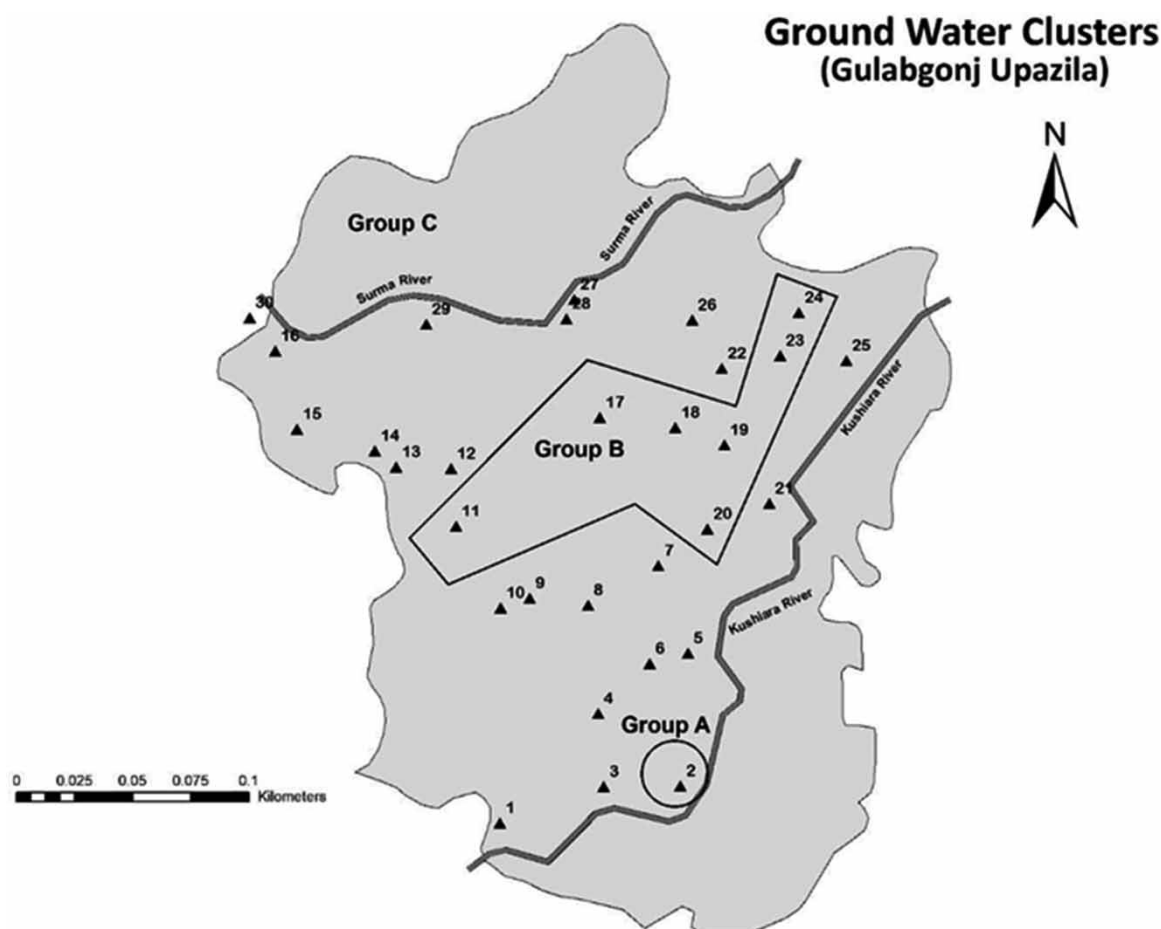


Figure 9 | Clusters of ground sample points at the site area.

elements and water quality parameters) and stepwise metal were used to construct discriminant functions, and the results of classification thus obtained are presented in [Table 12](#).

For surface water, the standard method has assigned the sample points 100% accurately to the disjointed groups generated from CA. However, for the same water, 96.7% of the assignments were correct, although each of the two discriminant functions were formed solely by iron and TDS. Similarly, the standard and stepwise methods correctly support the groups generated by CA for groundwater on an average of 76.7 and 70%, respectively. Each of the two discriminant functions contain manganese, zinc, nickel, and EC as the most significant parameters. Finally, the standard method and the stepwise method with discriminant functions based on the significant parameters of cobalt, manganese, pH, and EC could correctly classify the natural sources of water, i.e., surface and groundwater, with 96.7 and 93.3% accuracy, respectively.

4. CONCLUSIONS

According to the findings, the concentrations of arsenic, manganese, and iron are not strictly maintaining the standard levels. The quality of groundwater is better than that of surface water. However, the surface and groundwater in the research region may be appropriate for domestic, drinking, agricultural, and industrial uses depending on the trace element concentrations and the water quality parameters. However, for these purposes, a proper study incorporating relevant water quality parameters is required.

Although a high iron concentration is not highly hazardous, it can make water turbid and disrupt the flow medium. A moderately acidic pH value (less than 5) is seen in 20% of groundwater samples. Other metrics of most water samples in the study region are within permitted limits. The extracted first three PCs cover almost 70% of the total variance in the surface water quality data set on trace elements. On the contrary, for groundwater, the same category of PCs covers about 81.21% of the total variance. Inflows from the Surma and Kushiara rivers are likely to contaminate both surface and groundwater. In addition to PCA, these contamination sources are

Table 12 | Classification matrices from discriminant analysis

			Predicted group membership (%)			
Water type: surface			A	B	C	Total
Standard method ^a	Original	A	4(100.0)	0	0	4
		B	0	9(100.0)	0	9
		C	0	0	17(100.0)	17
Stepwise method ^b	Original	A	4(100.0)	0	0	4
		B	0	9(100.0)	0	9
		C	0	1(5.9)	16(94.1)	17
			Predicted group membership (%)			
Water type: ground			A	B	C	Total
Standard methods ^c	Original	A	1(100.0)	0	0	1
		B	0	6(85.714)	1(14.286)	7
		C	0	0	22(100.0)	22
Stepwise method ^d	Original	A	1(100.0)	0	0	1
		B	0	7(100.0)	0	7
		C	0	3(13.6)	19(86.4)	22
			Predicted group membership (%)			
Water type: surface (S) and ground (G)			Surface	Ground		Total
Standard method ^e	Original	S	29(96.7)	1(3.3)		30
		G	1(3.3)	29(96.7)		30
Stepwise method ^f	Original	S	28(93.3)	2(6.7)		30
		G	2(6.7)	28(93.3)		30

^a100.0% of the original grouped cases were correctly classified.^b96.7% of the original grouped cases were correctly classified.^c96.7% of the original grouped cases were correctly classified.^d90.0% of the original grouped cases were correctly classified.^e96.7% of the original grouped cases were correctly classified.^f93.3% of the original grouped cases were correctly classified.

supported by CA, which produced the largest cluster for both surface and groundwater located close to either of the rivers. Thus, policymakers may take action to mitigate the deterioration of the water quality of the Surma and Kushiara rivers, which are mostly contaminated by the activities in their catchment areas.

The classification of surface water as well as groundwater and the natural difference between surface and groundwater quality are strongly supported by the DA. Soil and rock-forming minerals were not included in the study, and these minerals can classify the study region in an alternative way. Around 40% of the surface water sample and 30% of the groundwater sample exceed the manganese permissible level. This condition may also be linked to an elevated level of iron or arsenic, sulfur-like odors, and hardness issues. Arsenic and manganese-rich water can be hazardous and carcinogenic.

Therefore, it is necessary to mark and prohibit drinking from water sources that are high in arsenic and manganese. Iron, a prevalent element, may also stall the manufacturing capacity of an industry through its grave. Therefore, alternate water sources might be used. Waters that are intended for drinking must be adequately treated to prevent health problems. To promote sustainable development, policymakers should concentrate more emphasis on their source of production, as it is more effective to avoid the generation of harmful elements than to remedy human-caused contaminations.

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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First received 8 June 2022; accepted in revised form 15 October 2022. Available online 31 October 2022